SHRAVAN SURESH KATKAR

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PROFESSIONAL SUMMARY

Aspiring AI Intern and B.Sc. Computer Science undergraduate (expected 2027) with hands-on project experience in Python, machine learning, and Generative AI. Skilled in building Retrieval-Augmented Generation (RAG) applications, document processing workflows, local LLM integrations, and computer vision solutions. Experienced with FastAPI, LangChain, ChromaDB, Ollama, REST APIs, JSON, and Pytest. Demonstrates analytical problem-solving through practical AI development, retrieval evaluation, and backend testing. Enthusiastic about learning and applying AI technologies to real-world engineering challenges.

TECHNICAL SKILLS

Programming: Python, C, HTML, CSS
Artificial Intelligence: Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), RAG, NLP, Computer Vision
Frameworks and Libraries: FastAPI, LangChain, YOLOv8, BLIP, Ollama
Backend and Databases: REST APIs, WebSockets, JSON, ChromaDB, SQLite, MySQL
Development and Testing: Git, GitHub, Pytest, Postman, Regression Testing

PROFESSIONAL EXPERIENCE

VaultAI – Personal Project *[Feb 2026 – March 2026]*
Offline RAG Document Intelligence Platform | Python, FastAPI, LangChain, ChromaDB, Ollama, React

- Engineered an offline document intelligence application with local LLM inference, document ingestion, text processing, embedding generation, and persistent vector search.
- Evaluated hybrid retrieval using 65 questions across 3 runs, completing 195 retrieval evaluations with Reciprocal Rank Fusion (RRF) and reranking.
- Recorded 0.9538 Hit@1 and 0.9769 Mean Reciprocal Rank (MRR) in the supplied retrieval benchmark.
- Resolved backend API reliability defects and expanded automated regression coverage; the recorded Phase 2 test run passed 26/26 tests.

ALIAS – Personal Project *[Oct 2025 – Dec 2025]*
AI Desktop Assistant | Python, FastAPI, Electron, LangChain, SQLite, WebSockets

- Built a Windows desktop AI assistant connecting multiple LLM providers with streamed responses and persistent conversation history.
- Integrated document-based RAG, speech interaction, and desktop automation through a Python backend and Electron interface.

CivicMind AI – Personal Project *[Jan 2026 – Feb 2026]*
Worker Safety Intelligence Platform | Python, FastAPI, React, YOLOv8, BLIP, SQLite

- Combined YOLOv8 object detection and BLIP scene understanding with a deterministic risk-scoring engine for workplace hazard assessment.
- Designed location-aware risk reporting, multilingual alerts, and a supervisor dashboard for field monitoring.

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science **8.73 CGPA**

JSPM's Rajarshi Shahu Commerce and Science College, Pune
Savitribai Phule Pune University (SPPU) Expected Graduation: 2027

Higher Secondary Certificate (HSC) **76.17%**

Shree Samarth Uchha Madhyamik Vidyalay, Pune

CERTIFICATIONS AND ACTIVITIES

- Complete Generative AI Course with LangChain and Hugging Face – Udemy.
- SQL (Advanced) – HackerRank.
- Hack The Horizon 2.0 – 24-hour hackathon participant, AI Club, VIT Chennai and SW VIT.